Risk and Uncertainty

So far in our models: decisions $\xrightarrow{\text{direct}}$ outcomes

many real world decisions involve risk

→ what makes a situation "risky"?
- uncertainty
- bad outcome
- spread in outcomes

how do we model risky situations

1) outcomes: the things that could happen
2) probabilities: how likely different values are

} a "lottery"

ex. you buy a bike worth \$400. There is a 25% chance that it is stolen

	outcome	value	probability
A.	not stolen	\$400	0.75
B.	stolen	\$0	0.25

Notes: probabilities are numbers between zero and one
- add up to 1
- probability = 1 : certainty
- probability = 0 : never happens

How do we measure risk?

Expected Value: on average, what do you expect to get?

$$EV = \begin{pmatrix}\text{probability}\\ \text{of } A\end{pmatrix} \cdot \begin{pmatrix}\text{value}\\ \text{of } A\end{pmatrix} + \begin{pmatrix}\text{probability}\\ \text{of } B\end{pmatrix} \cdot \begin{pmatrix}\text{value}\\ \text{of } B\end{pmatrix}$$

ex. $EV = (0.75)\cdot(400) + (0.25)\cdot(0) = 300$

Variance: a measure of spread

$$Var = \begin{pmatrix}\text{probability}\\ \text{of } A\end{pmatrix}\cdot\begin{pmatrix}\text{difference in}\\ A \text{ and } EV\end{pmatrix}^2 + \begin{pmatrix}\text{probability}\\ \text{of } B\end{pmatrix}\cdot\begin{pmatrix}\text{difference in}\\ B \text{ and } EV\end{pmatrix}^2$$

$$V(x) = E(x^2) - E(x)^2$$

Standard Deviation $= \sqrt{\text{variance}}$

But, how do we model choices between risky alternatives

Expected Utility: the utility of each outcome weighted by probabilities

$$EU = \begin{pmatrix}\text{probability}\\ \text{of } A\end{pmatrix}\cdot U(A) + \begin{pmatrix}\text{probability}\\ \text{of } B\end{pmatrix}\cdot U(B)$$

choosing between risky alternatives

consider the following lottery: I flip a coin → heads: you get paid \$4
↳ tails: you get paid \$0

$EV = (\frac{1}{2})(4) + (\frac{1}{2})(0) = 2$

$Var = (\frac{1}{2})(4-2)^2 + (\frac{1}{2})(0-2)^2 = 4$

$SD = \sqrt{4} = 2$

Attitudes towards risk

1) Risk Aversion ex. utility function over outcome X $U = X^{1/2}$

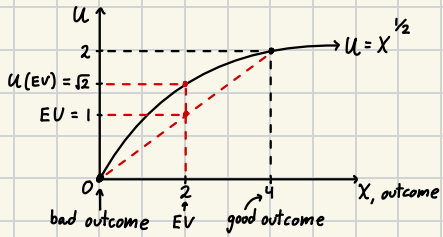
Expected Utility: $EU = (\frac{1}{2})U(4) + (\frac{1}{2})U(0) = (\frac{1}{2})(2) + (\frac{1}{2})(0) = 1$

What if I just gave this person \$2 (EV) instead

Utility of Expected Value $= U(EV) = 2^{1/2} = 1.414\ldots$

Notice that $U(EV) > EU$ for this person

- this will always be true for someone who is risk averse



a risk averse person would prefer to give up the chance of getting \$4 and EV for sure

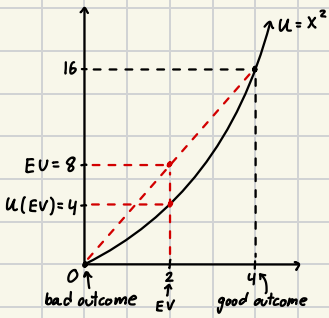
2) Risk Loving Preferences ex. utility function $U = X^2$

same lottery as before → heads: \$4 → $U(4) = 16$
↳ tails: \$0 → $U(0) = 0$

expected utility: $EU = (\frac{1}{2})(16) + (\frac{1}{2})(0) = 8$

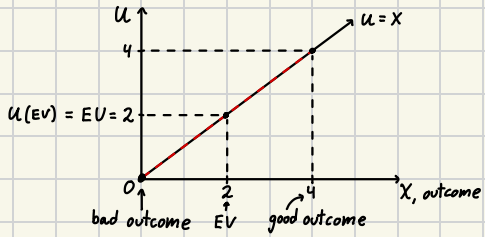
if we gave this person the EV with certainty: $U(EV) = U(2) = 4$

so for this person $EU > U(EV)$



a risk loving person prefers the lottery over getting EV for sure

3) Risk Neutral ex. utility function $u = x$



a risk neutral person only cares about EV (not variance)